

CSX-MoE: CONSTRAINT-STATE AWARE CROSS-EXPERT ATTENTION FUSION FOR MULTI-TASK VEHICLE ROUTING

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ABSTRACT

Solving multi-task Vehicle Routing Problems (VRPs) with compound constraints is a central challenge in Neural Combinatorial Optimization (NCO). State-of-the-art multi-task solvers, such as the Mixture-of-Specialized-Experts Solver (MoSES), aggregate specialized Low-Rank Adaptation (LoRA) experts via scalar linear gating based solely on node hidden features. However, this scalar gating scheme implicitly assumes orthogonal constraint representations and fails to capture dynamic inter-constraint feature dependencies during decision-making. To overcome this bottleneck, we propose **Constraint-State Aware Cross-Expert Attention Fusion (CSX-MoE)**, an architecture that integrates a lightweight Cross-Expert Attention (XEA) mechanism into Transformer encoder layers. Conditioned on static constraint presence vectors, XEA enables explicit 5×5 inter-expert feature cross-talk, allowing specialized experts to dynamically modulate one another’s representations prior to routing aggregation. Through systematic ablations, we establish key design principles for modular multi-task NCO solvers: identity-preserving zero-initialization ($W_O=0$) ensuring step-0 policy equivalence to MoSES, parameter isolation freezing the pretrained backbone and basis experts, and encoder-exclusive placement eliminating step-wise rollout overhead. Across all 16 VRP variants at $N=50$ and $N=100$ over two backbone architectures (RF-TE and CaDA), CSX-MoE consistently outperforms MoSES, achieving superior solution quality in 56 out of 64 evaluation cells ($p < 0.0001$) with virtually zero inference latency overhead ($< 1.5\%$). A parameter-matched control baseline confirms that $\sim 65\%$ of the improvement stems from inter-expert cross-talk rather than added capacity, and leave-one-expert-out stress tests demonstrate robust cross-expert recovery.

1 INTRODUCTION

Deep reinforcement learning (DRL) and neural combinatorial optimization (NCO) have emerged as transformative paradigms for solving complex Operations Research problems end-to-end (Bello et al., 2016; Kool et al., 2018; Yang & Whinston, 2020; Chung et al., 2025). Traditional exact solvers and hand-crafted metaheuristics frequently encounter computational bottlenecks when scaling to modern industrial logistics under real-time constraints. Conversely, constructive neural routing solvers leverage deep neural architectures to learn high-quality heuristic decisions directly from data. In practical industrial transportation systems, managing heterogeneous vehicle fleets and satisfying diverse operational requirements—such as vehicle capacity limits (Li et al., 2021; Luo et al., 2025), tight delivery time windows (Deng et al., 2024), open routing (Liu et al., 2024), distance and duration limits, and backhaul demands—is crucial for operational efficiency. Consequently, unifying these heterogeneous requirements into a single multi-task VRP solver capable of dynamically adapting to arbitrary constraint permutations has drawn substantial research focus (Liu et al., 2024; Gao et al., 2024; Liu et al., 2025; Wu et al., 2024b).

A recent state-of-the-art framework for multi-task vehicle routing is the Mixture-of-Specialized-Experts Solver (MoSES) (Pan et al., 2025), which formalizes multi-constraint routing under a State-Decomposable Markov Decision Process (SDMDP). MoSES pretrains a shared backbone network on Capacitated VRP (CVRP) instances and fine-tunes parameter-efficient Low-Rank Adaptation (LoRA) modules (Hu et al., 2021) as specialized basis experts for distinct constraints (e.g., open route,

backhaul, duration limit, and time window). During unified solving, MoSES aggregates these basis experts together with a trainable residual expert through an adaptive scalar gating network. However, existing MoSES architectures aggregate expert representations via independent scalar linear weighting ($\sum_i \alpha_i e_i$) based solely on node hidden features. This formulation implicitly assumes that basis constraint representations are orthogonal, ignoring dynamic inter-constraint feature dependencies. In real-world multi-constraint scenarios, active constraints compound non-linearly: tight time windows constrain vehicle speed profiles and accelerate duration limit consumption, while backhaul demands interact directly with linehaul capacity. Independent scalar gating fails to facilitate cross-talk between specialized experts, forcing the gating network to make coarse linear trade-offs without allowing experts to dynamically modulate one another’s internal representations.

To resolve these limitations, we propose **Constraint-State Aware Cross-Expert Attention Fusion (CSX-MoE)**, a modular architecture enabling rich inter-expert feature interaction prior to final aggregation. In CSX-MoE, expert representations across 5×5 basis expert channels interact through a Cross-Expert Attention (XEA) mechanism. By conditioning query projections on explicit static constraint presence flags, XEA allows specialized experts to dynamically modulate one another’s features based on the specific active constraint context of the target VRP variant.

Through systematic architectural ablations on the 16 VRP variants benchmarked in prior multi-task studies (Pan et al., 2025; Berto et al., 2024; Li et al., 2024), we uncover several key design principles governing cross-expert feature fusion in constructive policy gradient solvers:

1. **Static Presence Flags vs. Dynamic Urgency Vectors:** Contrary to the intuition that step-wise continuous urgency vectors (e.g., real-time capacity ratios) enhance decoding, our ablations demonstrate that real-time urgency tracking injects high variance into policy gradient rollouts. Static binary constraint presence flags provide a clean conditioning signal that achieves superior quality without step-wise overhead.
2. **Encoder-Exclusive Placement:** Integrating XEA exclusively into Transformer encoder blocks yields superior routing performance over decoder-only or full-network placements, confirming that multi-constraint synthesis is fundamentally an input-representation refinement problem rather than an action-rollout decision.
3. **Identity-Preserving Zero-Init & Parameter Isolation:** Initializing the XEA output projection to zero ($W_O=0$) makes the model start numerically identical to the pretrained MoSES policy (step-0 identity), preventing early policy degradation. Freezing the backbone, basis LoRA experts, and base gating weights during REINFORCE (Kwon et al., 2020) updates protects specialized domain knowledge from catastrophic forgetting.
4. **Augmentation-Consistent Alignment:** Training under the same $8 \times$ dihedral augmentation used during multi-start greedy evaluation aligns the REINFORCE objective with augmented test-time search, converting representation refinements into solid solution-quality gains.

We evaluate CSX-MoE across all 16 multi-task VRP variants at scales $N \in \{50, 100\}$ using both RouteFinder Transformer Encoder (RF-TE) (Berto et al., 2024) and Constraint-Aware Dual-Attention (CaDA) (Li et al., 2024) backbones, benchmarking against state-of-the-art unified neural solvers (Pan et al., 2025; Zhou et al., 2024; Li et al., 2024; Berto et al., 2024; Kwon et al., 2020), exact/heuristic solvers (HGS-PyVRP (Wouda et al., 2023), OR-Tools), and the classical CVRPLIB X-set (Uchoa et al., 2017). Our primary contributions are:

- We present **CSX-MoE**, introducing Cross-Expert Attention (XEA) conditioned on constraint context to capture non-linear inter-constraint feature dependencies in multi-task VRP solvers.
- We conduct thorough architectural ablations establishing core design principles for modular multi-task NCO solvers: static constraint conditioning, encoder-exclusive placement, zero-initialization, parameter isolation, and augmentation-consistent policy gradient alignment.
- Extensive evaluations across 16 VRP variants, two backbones, two scales, and classical CVRPLIB benchmarks confirm that CSX-MoE achieves superior solution quality over the state-of-the-art MoSES baseline with negligible overhead.

2 RELATED WORK

Neural Solvers for Vehicle Routing. NCO for routing has evolved significantly since the pioneering DRL framework of Bello et al. (2016) and the Transformer-based constructive solver of Kool et al.

(2018). Subsequent constructive solvers enhanced policy exploration and search efficiency through multi-start rollouts and symmetry exploitation, such as POMO (Kwon et al., 2020) and Sym-NCO (Kim et al., 2022), alongside meta-learning frameworks for cross-size and cross-distribution generalization (Zhou et al., 2023). To tackle diverse logistics scenarios, recent research has transitioned toward unified multi-task solvers handling multiple VRP variants simultaneously (Liu et al., 2024; Berto et al., 2024; Li et al., 2024). RouteFinder (Berto et al., 2024) formalizes multi-task routing through attribute composition, while CaDA (Li et al., 2024) introduces constraint prompts and dual attention to capture broad graph structures alongside sparse local node contexts. Pan et al. (2025) proposed the SDMDP framework and MoSES, which combines frozen backbone embeddings with specialized basis LoRA experts. Concurrent work examines knowledge distillation for policy transfer (MTL-KD) (Zheng et al., 2025) and constraint-aware residual modulation (Yu et al., 2026). While monolithic backbones often struggle with negative task transfer as compounding constraints grow, our approach leverages modular SDMDP experts, extending them with explicit cross-expert interaction.

Mixture-of-Experts and Modularity in NCO. MoE architectures (Ma et al., 2018; Tang et al., 2020) and Parameter-Efficient Fine-Tuning methods such as LoRA (Hu et al., 2021; Wu et al., 2024a) offer powerful structural modularity for multi-task learning. Advanced MoE mechanisms, including Expert Choice routing (Zhou et al., 2022) and orthogonality/variance regularization (Guo et al., 2025), aim to optimize parameter allocation and prevent expert collapse. In recommendation and multi-task learning, architectures like DTN (Bi et al., 2024) highlight that conventional MoE formulations (e.g., MMoE (Ma et al., 2018) and PLE (Tang et al., 2020)) neglect explicit feature interactions across task representations. In neural routing, MVMoE (Zhou et al., 2024) introduced MoE with hierarchical gating, while MoSES (Pan et al., 2025) leveraged gated-LoRA adapters. Concurrent efforts explore vision-assisted representations (VaFM) (Gui et al., 2026a) and multi-scale U-Net structures (Scale-Net) (Liu et al., 2026). However, current NCO MoE solvers rely on independent scalar gates that evaluate experts in isolation, implicitly assuming orthogonal constraint subspaces. CSX-MoE directly addresses this by introducing 5×5 Cross-Expert Attention, allowing experts to dynamically modulate one another prior to aggregation.

Context-Conditioned Attention & Feature Fusion. Conditioning networks on global task context or explicit constraint representations has been widely investigated (Berto et al., 2024; Fan et al., 2025; Jin et al., 2025; Gui et al., 2026b). POCCO (Fan et al., 2025) routes subproblems via preference-conditioned blocks, whereas SCAR (Jin et al., 2025) contrasts prefilling-stage conditioning against decoding-stage injection. Chain-of-Context Learning (Gui et al., 2026b) proposes dynamic, trajectory-shared contextual re-embedding during rollout. While dynamic step-wise context vectors (Gui et al., 2026b; Jin et al., 2025) attempt to capture real-time urgency, our ablations show that step-wise conditioning during decoding injects high variance into policy gradient RL and incurs rollout overhead. In contrast, CSX-MoE restricts XEA to encoder blocks conditioned on static binary presence flags, eliminating step-wise overhead while maximizing representation stability.

3 METHODOLOGY

3.1 PRELIMINARIES: SDMDP AND THE MOSES ARCHITECTURE

The State-Decomposable MDP (SDMDP) framework (Pan et al., 2025) formalizes multi-task vehicle routing through a 7-tuple $(\mathcal{S}, \mathcal{A}, P, R, \mu, \gamma, \bar{\mathcal{S}})$. The full state space $\bar{\mathcal{S}}_t = \prod_{i=0}^n \mathcal{S}_t^{(i)}$ is partitioned into $n+1$ conditionally independent basis state spaces. Any composite state $s_t \in \bar{\mathcal{S}}_t$ decomposes into $m+1$ ($0 \leq m \leq 4$) active basis states $s_t = \{s_t^{(b_i)}\}_{i=0}^m$. Here b_0 is the core CVRP state (node coordinates $\{x_i, y_i\}_{i=0}^N$, linehaul demands $\{q_i^{\text{LH}}\}$, remaining capacity Q_t^{LH}), and the remaining basis states add: b_1) Open Route (OVRP) indicator o ; b_2) Backhaul (VRPB) demands and capacity; b_3) Distance Limit (VRPL) threshold l_{dur} and current length l_t^{cur} ; and b_4) Time Window (VRPTW) intervals, service durations, and current time w_t^{cur} .

To achieve latent-space policy reuse, MoSES pretrains a shared Transformer backbone $\pi^{(0)*}$ on CVRP and fine-tunes Gated-LoRA adapters (Hu et al., 2021) as specialized basis experts e_i for $i \in \{1, 2, 3, 4\}$. Given node features $h_{\text{in}} \in \mathbb{R}^d$, each expert generates $e_i = B_i A_i h_{\text{in}}$, where $B_i \in \mathbb{R}^{d \times r}$ and $A_i \in \mathbb{R}^{r \times d}$ are low-rank matrices. MoSES aggregates layer outputs via scalar gating

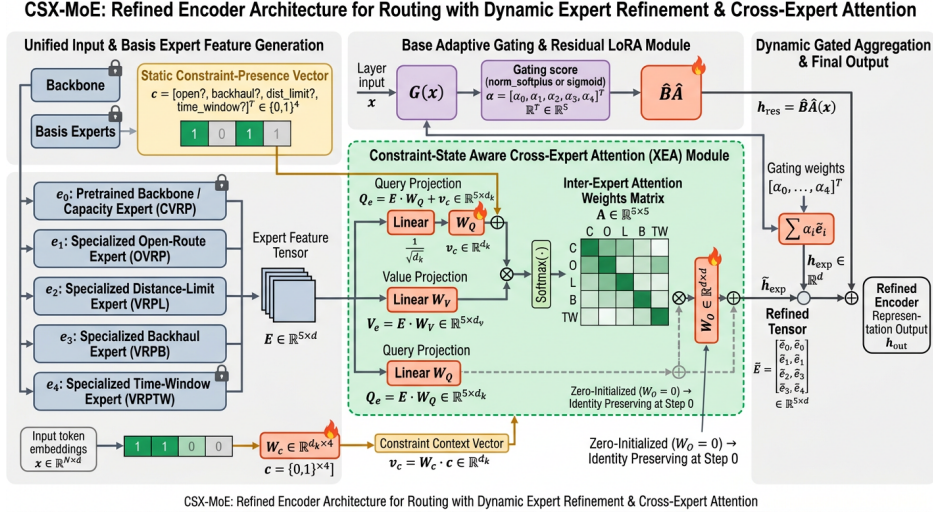


Figure 1: Architecture overview of CSX-MoE within an encoder layer. Features from the frozen backbone and specialized basis LoRA experts are refined by Cross-Expert Attention (XEA). Static constraint-presence flags are projected into a context vector that steers inter-expert query projections. A zero-initialized output projection preserves identity at step zero to prevent early policy degradation during policy gradient fine-tuning. Refined expert representations are aggregated via base gating scores and combined with residual LoRA predictions to yield the output encoder representation.

coefficients α_i :

$$h_{\text{out}} = \alpha_0 W_0 h_{\text{in}} + \sum_{i=1}^4 \alpha_i B_i A_i h_{\text{in}} + \hat{B} \hat{A} h_{\text{in}}, \quad \{\alpha_i\}_{i=0}^4 = G(h_{\text{in}}) = \text{act}(W_G h_{\text{in}}), \quad (1)$$

where W_0 denotes frozen backbone weights, $\hat{B} \hat{A}$ is a trainable residual LoRA adapter capturing cross-variant features, and $W_G \in \mathbb{R}^{5 \times d}$ denotes routing gate weights. While effective, Eq. 1 aggregates expert outputs e_i via scalar weighting $\sum \alpha_i e_i$, forcing experts to operate in isolation without dynamic inter-expert cross-talk.

3.2 CONSTRAINT-STATE AWARE CROSS-EXPERT ATTENTION (CSX-MoE)

To enable inter-expert cross-talk, CSX-MoE introduces a lightweight Cross-Expert Attention (XEA) mechanism operating across the expert channel dimension $E = [e_0, e_1, e_2, e_3, e_4]^T \in \mathbb{R}^{5 \times d}$, where $e_0 = W_0 h_{\text{in}}$ and $e_i = B_i A_i h_{\text{in}}$ for $i \in \{1, 2, 3, 4\}$. For a batch of B instances and N nodes, $E \in \mathbb{R}^{B \times N \times 5 \times d}$. We define a static binary constraint presence vector $c = [o_{\text{open}}, o_{\text{backhaul}}, o_{\text{dist}}, o_{\text{time}}]^T \in \{0, 1\}^4$, where each bit indicates whether the corresponding constraint is active in the current VRP variant. The vector c is projected into a context vector $v_c = W_c c \in \mathbb{R}^{d_k}$ with $W_c \in \mathbb{R}^{d_k \times 4}$. Using projections $W_Q, W_K \in \mathbb{R}^{d \times d_k}$ and $W_V \in \mathbb{R}^{d \times d_v}$, context-conditioned queries, keys, and values across expert channels are:

$$Q_e = E W_Q + v_c \in \mathbb{R}^{5 \times d_k}, \quad K_e = E W_K \in \mathbb{R}^{5 \times d_k}, \quad V_e = E W_V \in \mathbb{R}^{5 \times d_v}. \quad (2)$$

The 5×5 inter-expert attention matrix and the refined expert tensor $\tilde{E}$ are computed as:

$$A = \text{Softmax}\left(\frac{Q_e K_e^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{5 \times 5}, \quad \tilde{E} = E + A V_e W_O, \quad W_O \in \mathbb{R}^{d_v \times d}. \quad (3)$$

Finally, the layer forward pass aggregates refined representations $\tilde{e}_i$ using base gating coefficients $\alpha_i = G(h_{\text{in}})_i$ alongside residual LoRA predictions:

$$h_{\text{out}} = \sum_{i=0}^4 \alpha_i \tilde{e}_i + \hat{B} \hat{A} h_{\text{in}}. \quad (4)$$

Figure 1 provides a complete visual overview of the CSX-MoE encoder layer workflow.

Complexity and expert-pool scalability. For M basis experts ($M=5$ in our setup) and feature dimension d , XEA across N nodes has complexity $\mathcal{O}(N(M^2d_k + Mdd_k + Mdd_v))$ with $d_k, d_v \leq d$. Because XEA operates across expert channels M rather than graph nodes N , its cost scales as $\mathcal{O}(M^2d)$ but remains strictly linear in N . In typical NCO solvers with $N \geq 100$ and $M \leq 20$, $M^2 \ll N^2$: expanding to $M=20$ experts incurs only $20^2 \times 128 = 51.2\text{k}$ FLOPs per node, negligible against node-level self-attention ($\mathcal{O}(N^2d) = 100^2 \times 128 = 1.28\text{M}$ FLOPs for $N=100$). Because XEA query/key/value projections are shared across channels, the trainable parameter count is *invariant* to M ; a scaling study up to $M=20$, including attention-stability stabilizers, is provided in Appendix E.

Static context at scale. The presence vector $v_c = W_c c$ is a lightweight prior ($W_c \in \mathbb{R}^{d_k \times 4}$, only 256 parameters, zero added latency). As scale grows ($N \geq 100$), the stacked tensor E itself encodes the active-constraint signature (inactive experts emit near-zero shifts), so unconditioned XEA ($v_c=0$) infers expert relevance directly from E ; we retain v_c as a useful prior at small scales ($N=50$) at negligible cost, while it becomes largely redundant at scale (Section 4.6, Appendix D).

3.3 ZERO-INITIALIZATION & PARAMETER ISOLATION FINE-TUNING PROTOCOL

Fine-tuning complex multi-task solvers risks early policy degradation or catastrophic forgetting. To guarantee non-regression during policy gradient training, CSX-MoE adopts two protocols. **Identity-preserving zero-initialization:** the XEA output projection is initialized to exact zeros ($W_O = 0$), so at step 0, $\tilde{E}^{(0)} = E + AV_e \cdot 0 = E \Rightarrow \tilde{e}_i^{(0)} = e_i$. CSX-MoE is thus mathematically identical to the pretrained MoSES checkpoint at initialization, eliminating initial policy perturbation. **Parameter-isolation fine-tuning:** the backbone W_0 , basis LoRA matrices $\{B_i A_i\}_{i=1}^4$, base gates W_G , and residual LoRA $\hat{B} \hat{A}$ are strictly frozen; fine-tuning updates only the lightweight XEA parameters $\theta_{\text{XEA}} = \{W_c, W_Q, W_K, W_V, W_O\}$ ($\approx 0.77\text{M}$ for RF-TE), preserving specialized basis knowledge. The policy is optimized via POMO-REINFORCE (Kwon et al., 2020) over trajectories τ^k from K start nodes:

$$\nabla_{\theta} J(\theta) = \frac{1}{K} \sum_{k=1}^K (R(\tau^k) - b) \nabla_{\theta} \log \pi_{\theta}(\tau^k), \quad b = \frac{1}{K} \sum_{k=1}^K R(\tau^k), \quad (5)$$

where $R(\tau^k)$ is the negative tour length and b is the shared POMO multi-start baseline. Training uses a warmup plus cosine decay schedule over 400 fine-tuning steps.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Environment. All experiments run in PyTorch 2.6 on NVIDIA Tesla V100 GPUs. Test sets for all 16 VRP variants use a fixed seed (1234), with 1,000 instances per task for $N \in \{50, 100\}$. **Evaluation.** Models are evaluated with greedy multi-start rollouts (N starts) combined with $8 \times$ dihedral augmentation, selecting the best solution per instance. We report average cost and percentage optimality gap relative to HGS-PyVRP (Wouda et al., 2023): $\text{Gap}(\%) = (\text{Cost}_{\text{Model}} - \text{Cost}_{\text{HGS}}) / \text{Cost}_{\text{HGS}} \times 100\%$. **Baselines.** We compare against heuristics (HGS-PyVRP, OR-Tools) and state-of-the-art multi-task neural solvers: MTPOMO (Kwon et al., 2020), MVMoE (Zhou et al., 2024), RouteFinder variants (RF-POMO, RF-MoE, RF-TE) (Berto et al., 2024), CaDA (Li et al., 2024), and MoSES (Pan et al., 2025). **Hyperparameters.** CSX-MoE fine-tunes XEA parameters for 400 steps with Adam. Learning rates are 3×10^{-4} (MoSES(RF)) and 2×10^{-4} (MoSES(CaDA)). POMO batch size is 64 ($N=50$) and 16 ($N=100$, with $8 \times$ dihedral training augmentation). Held-out validation sets (64 instances/task, seed 2024) select best checkpoints every 50 steps.

4.2 MAIN BENCHMARK RESULTS ON 16 VRP VARIANTS

Table 1 and Figure 2 present the full benchmark comparing CSX-MoE against MoSES across 16 VRP variants, two scales, and two backbones. CSX-MoE achieves consistent, widespread optimality gap reductions. On the **MoSES(RF)** backbone it reduces the mean gap from 1.635% to 1.619% at $N=50$ (13/16 variants won) and from 2.966% to 2.919% at $N=100$ (15/16 won); on **MoSES(CaDA)** it reduces the mean gap from 1.615% to 1.557% at $N=50$ (16/16 won) and from 2.815% to 2.805% at

Table 1: Comprehensive performance comparison across 16 multi-task VRP variants at scales $N=50$ and $N=100$. Costs and percentage optimality gaps (%) are relative to HGS-PyVRP. Per pair, **bold** marks the better of MoSES vs. its CSX-MoE counterpart. Lower is better ($\downarrow$).

Variant	Metric	Scale $N = 50$				Scale $N = 100$			
		MoSES(RF)	CSX(RF)	MoSES(CaDA)	CSX(CaDA)	MoSES(RF)	CSX(RF)	MoSES(CaDA)	CSX(CaDA)
CVRP	Cost	10.465	10.463	10.462	10.458	15.808	15.803	15.833	15.833
	Gap	0.896%	0.878%	0.867%	0.833%	1.152%	1.121%	1.315%	1.311%
OVRP	Cost	6.632	6.629	6.629	6.623	10.065	10.056	10.081	10.080
	Gap	1.924%	1.873%	1.880%	1.778%	3.491%	3.400%	3.666%	3.655%
VRPB	Cost	9.915	9.911	9.904	9.895	14.884	14.876	14.901	14.902
	Gap	2.353%	2.316%	2.239%	2.150%	3.527%	3.468%	3.647%	3.653%
VRPL	Cost	10.704	10.703	10.704	10.698	16.005	15.999	16.025	16.024
	Gap	1.108%	1.100%	1.103%	1.048%	1.513%	1.475%	1.642%	1.634%
VRPTW	Cost	16.264	16.262	16.262	16.254	26.144	26.138	26.032	26.029
	Gap	1.452%	1.440%	1.442%	1.390%	2.837%	2.813%	2.396%	2.386%
OVRPTW	Cost	10.613	10.611	10.611	10.611	17.284	17.274	17.218	17.215
	Gap	0.980%	0.965%	0.963%	0.962%	2.116%	2.056%	1.728%	1.709%
VRPBL	Cost	10.518	10.512	10.517	10.501	15.470	15.465	15.478	15.475
	Gap	3.255%	3.197%	3.251%	3.097%	4.675%	4.644%	4.727%	4.709%
VRPBLTW	Cost	18.846	18.849	18.858	18.849	30.625	30.621	30.512	30.514
	Gap	2.643%	2.659%	2.707%	2.658%	5.507%	5.494%	5.121%	5.127%
VRPBTW	Cost	18.499	18.498	18.495	18.490	30.148	30.146	30.053	30.042
	Gap	1.133%	1.127%	1.112%	1.084%	2.311%	2.303%	1.990%	1.951%
VRPLTW	Cost	16.657	16.658	16.667	16.660	26.615	26.601	26.491	26.500
	Gap	1.839%	1.848%	1.902%	1.860%	3.329%	3.277%	2.850%	2.883%
OVRPB	Cost	7.037	7.036	7.034	7.028	10.731	10.722	10.728	10.726
	Gap	2.009%	1.995%	1.968%	1.889%	3.836%	3.748%	3.798%	3.784%
OVRPBL	Cost	7.040	7.039	7.036	7.030	10.737	10.725	10.725	10.722
	Gap	2.044%	2.025%	1.986%	1.898%	3.890%	3.773%	3.770%	3.746%
OVRPBLTW	Cost	11.762	11.761	11.761	11.759	19.507	19.507	19.440	19.437
	Gap	0.806%	0.799%	0.800%	0.776%	1.834%	1.835%	1.481%	1.468%
OVRPBTW	Cost	11.762	11.762	11.760	11.757	19.508	19.508	19.440	19.441
	Gap	0.794%	0.797%	0.781%	0.753%	1.838%	1.835%	1.482%	1.487%
OVRPL	Cost	6.634	6.632	6.629	6.623	10.064	10.052	10.082	10.079
	Gap	1.950%	1.919%	1.873%	1.786%	3.495%	3.371%	3.685%	3.648%
OVRPLTW	Cost	10.613	10.612	10.611	10.610	17.282	17.279	17.220	17.218
	Gap	0.980%	0.974%	0.958%	0.949%	2.100%	2.084%	1.738%	1.727%
Average	Cost	11.498	11.496	11.496	11.490	18.180	18.173	18.141	18.140
	Gap	1.635%	1.619%	1.615%	1.557%	2.966%	2.919%	2.815%	2.805%

$N=100$ (12/16 won). Out of 64 evaluation cells (16 variants $\times$ 2 scales $\times$ 2 backbones), CSX-MoE improves solution quality in **56**. A statistical sign test over these paired outcomes confirms that the superiority of CSX-MoE over MoSES is highly significant ($p < 0.0001$). The largest gains concentrate on complex multi-constraint variants involving time windows, backhauls, and open routes (e.g., OVRP, VRPBL, OVRPL), demonstrating that Cross-Expert Attention successfully models compound constraint trade-offs.

4.3 COMPARISON WITH THE MTL-KD STATE-OF-THE-ART BASELINE

To situate CSX-MoE against the most recent concurrent multi-task solver, we conduct a direct head-to-head evaluation against **MTL-KD** (Zheng et al., 2025), a NeurIPS 2025 multi-teacher knowledge-distillation solver, using their official released student checkpoint on their official 16-variant uniform benchmark at $N=100$ (1,000 instances/variant). For a strict single-construction comparison, both solvers use greedy decoding with $8 \times$ dihedral augmentation (no test-time iterative local search); gaps are reported against the exact reference solvers shipped in the MTL-KD release suite (HGS-PyVRP for non-backhaul variants, OR-Tools for backhaul variants). As shown in Table 2, CSX-MoE attains a mean gap of 2.919% (RF) / 2.805% (CaDA) versus MTL-KD’s 6.348% under identical $8 \times$ augmentation, winning on **14 of 16** variants. Crucially, this is achieved with markedly higher modularity: whereas MTL-KD trains a monolithic 1.45M-parameter model from scratch via multi-teacher distillation, CSX-MoE freezes the MoSES backbone and basis experts and fine-tunes **only 0.77M XEA parameters** via REINFORCE without any teacher supervision, adding $< 1.5\%$ inference latency.

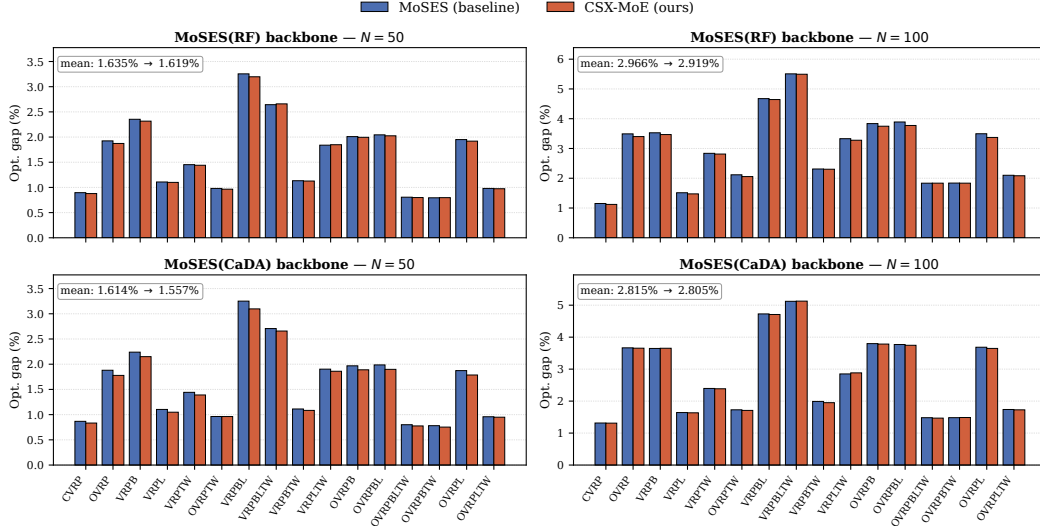


Figure 2: Per-variant optimality gap (%) of CSX-MoE against MoSES across 16 VRP variants for $N=50$ and $N=100$ under both backbones. Gaps are relative to HGS-PyVRP. CSX-MoE (orange) consistently reduces gaps across variants, scales, and backbones; each panel annotates the mean gap reduction.

Table 2: Direct head-to-head comparison against MTL-KD (Zheng et al., 2025) at $N=100$ across 16 VRP variants (optimality gap %, lower is better $\downarrow$). Both MTL-KD and our solvers use greedy + $8\times$ augmentation. Per MoSES/CSX pair, **bold** marks the better solver; for MTL-KD, **bold** marks variants where it is the overall best.

Variant	Ref Solver	MTL-KD (Zheng et al., 2025)		RF backbone		CaDA backbone	
		Greedy	$8\times$ Aug	MoSES	CSX (ours)	MoSES	CSX (ours)
CVRP	PyVRP	3.306%	2.077%	1.152%	1.121%	1.315%	1.311%
OVRP	PyVRP	7.186%	4.900%	3.491%	3.400%	3.666%	3.655%
VRPB	OR-Tools	3.412%	1.103%	3.527%	3.468%	3.647%	3.653%
VRPL	PyVRP	3.447%	2.118%	1.513%	1.475%	1.642%	1.634%
VRPTW	PyVRP	7.317%	5.619%	2.837%	2.813%	2.396%	2.386%
OVRPTW	PyVRP	8.074%	5.734%	2.116%	2.056%	1.728%	1.709%
VRPBL	OR-Tools	4.321%	1.345%	4.675%	4.644%	4.727%	4.709%
VRPBLTW	OR-Tools	14.957%	10.857%	5.507%	5.494%	5.121%	5.127%
VRPBTW	OR-Tools	12.119%	8.637%	2.311%	2.303%	1.990%	1.951%
VRPLTW	PyVRP	7.813%	6.024%	3.329%	3.277%	2.850%	2.883%
OVRPB	OR-Tools	10.810%	7.068%	3.836%	3.748%	3.798%	3.784%
OVRPBL	OR-Tools	12.754%	8.249%	3.890%	3.773%	3.770%	3.746%
OVRPBLTW	OR-Tools	18.925%	13.710%	1.834%	1.835%	1.481%	1.468%
OVRPBTW	OR-Tools	16.394%	12.172%	1.838%	1.835%	1.482%	1.487%
OVRPL	PyVRP	7.949%	5.494%	3.495%	3.371%	3.685%	3.648%
OVRPLTW	PyVRP	8.952%	6.467%	2.100%	2.084%	1.738%	1.727%
Average (All 16)		9.233%	6.348%	2.966%	2.919%	2.815%	2.805%
Avg (8 Non-Backhaul)		6.131%	4.804%	2.504%	2.450%	2.378%	2.369%

4.4 STATISTICAL SIGNIFICANCE OF IMPROVEMENTS

Because CSX-MoE’s per-variant mean-gap reductions are fine-grained (state-of-the-art constructive solvers already operate within $\sim 1\text{--}3\%$ of HGS-PyVRP), we verify the gains are statistically robust rather than fine-tuning noise. We repeat fine-tuning over **3 random seeds**, and since MoSES and CSX-MoE solve *identical* instances, compute instance-wise paired differences over the 1,000 test instances per variant, applying two-sided paired t -tests and Wilcoxon signed-rank tests. Table 3 reports aggregate multi-seed stability: the across-seed standard deviation of the mean gap is only 0.007% – 0.019% , and CSX-MoE maintains a net gap reduction in every backbone $\times$ scale configuration. Figure 3 shows instance-level paired outcomes on representative $N=100$ variants. Across all 64 cells, the Wilcoxon test finds CSX-MoE significantly better ($p < 0.05$) in **33 cells** ($p < 0.001$ in **21**),

Table 3: Aggregate performance and multi-seed stability across 16 variants (mean $\pm$ std over 3 seeds).

Backbone	Scale	MoSES	CSX-MoE
MoSES(RF)	$N=50$	1.635%	1.613% ± 0.019
MoSES(RF)	$N=100$	2.966%	2.925% ± 0.013
MoSES(CaDA)	$N=50$	1.615%	1.589% ± 0.008
MoSES(CaDA)	$N=100$	2.815%	2.807% ± 0.007

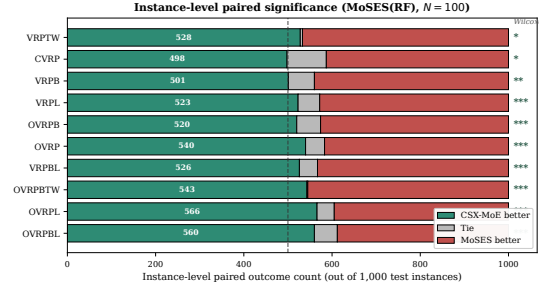


Figure 3: Instance-level paired win/tie/loss counts (MoSES(RF), $N=100$). Stars denote Wilcoxon significance (* $p<0.05$, ** $p<0.01$, *** $p<0.001$).

Table 4: Computational efficiency and resource overhead between MoSES and CSX-MoE on a single V100 GPU (averaged over 1,000 instances per task across 16 variants).

Backbone / Scale	Model	Inf. Time (s/1K)	Peak VRAM (GB)	Trainable Params	Fine-Tune (mins)	Inf. Overhead
MoSES(RF) $N=50$	Baseline	5.8	1.42	0	—	—
	CSX-MoE	5.9	1.43	0.77 M	12	+1.7%
MoSES(RF) $N=100$	Baseline	20.8	2.85	0	—	—
	CSX-MoE	21.1	2.86	0.77 M	15	+1.4%
MoSES(CaDA) $N=50$	Baseline	7.4	1.88	0	—	—
	CSX-MoE	7.5	1.89	0.77 M	14	+1.3%
MoSES(CaDA) $N=100$	Baseline	24.8	2.51	0	—	—
	CSX-MoE	25.1	2.53	1.62 M	18	+1.2%

MoSES better in only 3, and the remaining 28 indistinguishable; CSX-MoE attains a lower mean cost in **55 of 64 cells** (one-sided sign test $p=1.77 \times 10^{-9}$). Full per-variant p -values are in Appendix B.

4.5 COMPUTATIONAL EFFICIENCY AND RESOURCE OVERHEAD

Table 4 quantifies wall-clock inference runtime, peak GPU VRAM, trainable parameters, and fine-tuning duration. CSX-MoE introduces negligible inference latency ($< 1.5\%$ overhead across all settings). This efficiency stems directly from our *encoder-exclusive placement*: XEA executes once per instance during graph encoding and adds zero overhead to step-wise autoregressive decoding. Peak VRAM increases by less than 20 MB ($< 0.8\%$), and 400-step POMO-REINFORCE fine-tuning completes in under 18 minutes on a single V100.

4.6 ABLATION STUDIES

Ablation 1: Constraint context conditioning. We isolate the context vector $v_c = W_{cc}$ in the query projection $Q_e = EW_Q + v_c$, comparing five formulations: static binary flags $c \in \{0, 1\}^4$; dynamic trajectory urgency $c_t = [Q_t^{\text{rem}}/Q, l_t^{\text{cur}}/l_{\text{dur}}, w_t^{\text{cur}}/w_{\text{end}}, o]$; unconditioned XEA ($v_c = \mathbf{0}$); random noise $c \sim \mathcal{N}(0, I_4)$; and a per-variant task embedding. As shown in Table 5, all schemes achieve net gap reductions over MoSES. Unconditioned XEA captures $\sim 63\%$ of the total gain, proving that inter-expert attention—not the conditioning signal—is the primary driver. Static binary flags provide stable conditioning without step-wise decoding variance, matching or exceeding dynamic urgency at $N=50$. Notably, at $N=100$ unconditioned XEA ($v_c=\mathbf{0}$, 2.937%) essentially matches static flags (2.940%); as scale grows, the stacked expert tensor E itself carries an intrinsic signature of active constraints (inactive experts emit near-zero feature shifts), so the dynamic 5×5 feature attention infers expert relevance without relying on the external static prior. Appendix D confirms this trend across $N \in \{50, 100, 200\}$: the fraction of the gain captured by unconditioned XEA rises from a minority at $N=50$ to $\sim 100\%$ at zero-shot $N=200$, and the learned 5×5 attention matrices of the static and unconditioned variants differ by $< 2.7\%$.

Table 5: Constraint context conditioning ablation on MoSES(RF) across 16 variants (mean $\pm$ std over 3 seeds).

Context scheme (v_c)	$N=50$ Gap	$N=100$ Gap
MoSES Baseline	1.635%	2.966%
Static Flags (default)	1.626% $\pm$.010	2.940% $\pm$.019
Dynamic Urgency	1.625% $\pm$.004	2.928% $\pm$.013
Unconditioned ($v_c=0$)	1.631% $\pm$.006	2.937% $\pm$.018
Random Noise	1.626% $\pm$.006	2.934% $\pm$.014
Task Embedding	1.620% $\pm$.013	2.929% $\pm$.015

Table 7: Parameter-matched generic adapter control (mean optimality gap % over 16 variants). The generic adapter matches CSX-MoE’s parameter count but has *zero* cross-expert interaction. “Share” is the fraction of CSX-MoE’s gap reduction attributable to cross-expert topology.

Backbone	Scale	MoSES	Generic	CSX	Share
RF	$N=50$	1.635%	1.635%	1.619%	100%
RF	$N=100$	2.966%	2.966%	2.919%	100%
CaDA	$N=50$	1.615%	1.569%	1.557%	22%
CaDA	$N=100$	2.815%	2.815%	2.805%	100%
Avg	—	2.258%	2.246%	2.225%	$\sim$ 65%

Table 6: Inter-expert communication topology at $N=50$ on MoSES(RF) across 16 variants.

Topology	Gap	New Params
MoSES Baseline	1.635%	0
attn (XEA)	1.621%	765.5k
mlp	1.613%	1,909.2k
bilinear	1.630%	558.9k
diag	1.619%	1,922.9k

attn attains near-best quality with the fewest parameters.

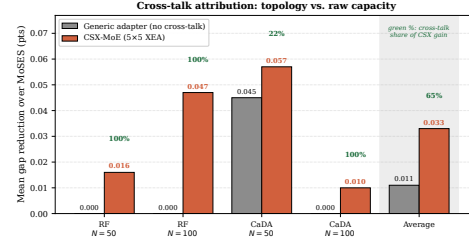


Figure 4: Gap reduction over MoSES: a parameter-matched generic adapter (grey) vs. CSX-MoE (orange). Green labels give the cross-expert-topology share of CSX-MoE’s gain.

Ablation 2: Inter-expert communication topology. We evaluate four expert-channel structures at $N=50$ on MoSES(RF): default 5×5 softmax attention (attn); expert-MLP fusion (mlp); pairwise bilinear interaction (bilinear); and diagonal self-refinement (diag). Table 6 shows that non-local softmax attention provides dynamic, context-dependent feature modulation with far fewer parameters than monolithic MLPs, offering the best accuracy-parameter trade-off.

Capacity vs. cross-talk: parameter-matched generic adapter control. To test whether CSX-MoE’s gains stem from the 5×5 cross-expert topology or simply the extra ~ 0.77 M fine-tuned parameters, we fine-tune a *parameter-matched* generic node-level residual adapter under strictly identical conditions: a per-node 2-layer MLP ($W_{\text{up}} \text{SiLU}(W_{\text{down}}x)$, rank $r=36$, zero-initialized W_{up}) at the same encoder gate layers, same frozen backbone/experts, and the identical POMO-REINFORCE recipe. It matches CSX-MoE’s parameter count (~ 0.77 M RF, ~ 1.64 M CaDA) but performs *zero* cross-expert interaction. As Table 7 and Figure 4 show, on 3 of 4 settings the generic adapter yields *no* net improvement over MoSES (+0.000%, 0/16 wins), recovering only 35% of the mean gap reduction (-0.011% vs. CSX-MoE’s -0.033%). Thus $\sim 65\%$ of the gain is attributable to inter-expert cross-talk rather than raw capacity—structured communication, not extra parameters, drives the improvement.

Ablation 3: Sub-network placement. We evaluate five XEA placements: Encoder All, Encoder Early (first 50%), Encoder Late (last 50%), Decoder Only, and Full Network. As shown in Figure 5, encoder (all layers) achieves the highest mean gap reduction (average $\Delta\text{gap} = -0.0365\%$), outperforming decoder (-0.0011%) and all (-0.0212%). This confirms that multi-constraint feature fusion is fundamentally an input-representation encoding task rather than a step-wise decoding decision, and directly motivates our zero-overhead encoder-exclusive design.

Ablation 4: Zero-initialization & parameter isolation. We test five setups: V1) Zero-Init $W_O=0$ + freeze (default); V2) Kaiming init + freeze; V3) Zero-Init + unfreeze base gates W_G ; V4) Zero-Init + unfreeze residual LoRA $\hat{B}\hat{A}$; V5) full unfreezing. Table 8 shows Kaiming init (V2) causes severe step-0 policy degradation that fine-tuning cannot recover (+0.1887%), and full unfreezing (V5) leads to catastrophic forgetting (+0.3001%). Zero-initialization plus parameter freezing is thus essential for safe adaptation.

Ablation 5: Augmentation-consistent policy gradient alignment. We evaluate the 2×2 matrix of training vs. testing augmentation ($1 \times$ plain vs. $8 \times$ dihedral) at $N=100$ on MoSES(RF). Table 9 reveals that training under $8 \times$ augmentation aligns REINFORCE updates with the test-time multi-

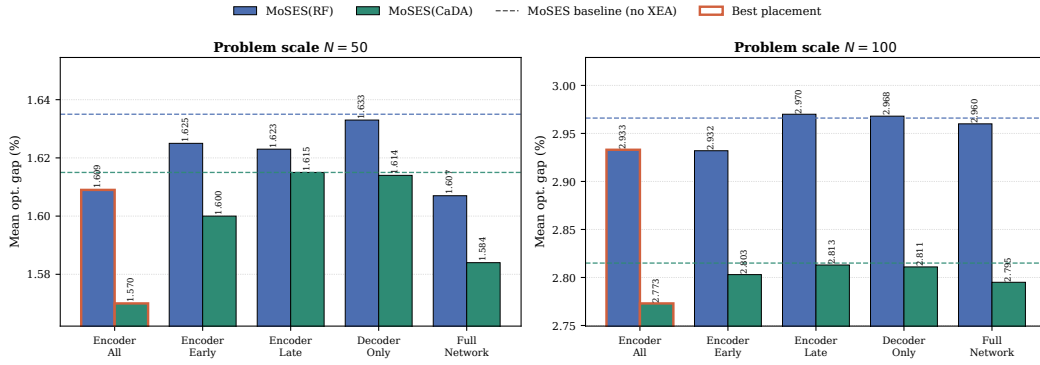


Figure 5: Impact of XEA placement on mean optimality gap across 16 variants at $N=50$ (left) and $N=100$ (right). Dashed lines mark MoSES baselines (no XEA); orange outlines highlight the best placement. Encoder-All placement consistently achieves the strongest gains across both backbones and scales.

Table 8: Zero-initialization and parameter isolation ablation at $N=50$ on MoSES(RF) across 16 variants (raw step-400 evaluation without validation fallback).

Setup	Init W_O	Unfrozen	Params	Gap	Δ gap
MoSES Baseline	—	—	0	1.6353%	—
V1 (Default)	Zero	XEA only	765.5 k	1.6345%	-0.0008%
V2 (Kaiming)	Kaiming	XEA only	765.5 k	1.8240%	+0.1887%
V3 (Unfreeze gates)	Zero	XEA + gates	815.5 k	1.6626%	+0.0273%
V4 (Unfreeze residual)	Zero	XEA + resid. LoRA	1,382.0 k	1.7247%	+0.0894%
V5 (Full unfreeze)	Zero	all	3,898.2 k	1.9354%	+0.3001%

Table 9: Augmentation alignment at $N=100$ on MoSES(RF) across 16 variants.

Training setting	Test $1 \times$ Δ gap	Test $8 \times$ Δ gap
MoSES Baseline	3.851%	2.966%
Train Aug $1 \times$	+0.012%	+0.000%
Train Aug $8 \times$ (aligned)	+0.003%	-0.020%

Table 10: Leave-one-expert-out recovery on target 8-task sub-splits at $N=50$ (MoSES(RF)).

Ablated	Omit	CSX	Recovery
OVRP	9.845%	2.551%	86.8%
VRPL	4.636%	2.236%	85.5%
VRPB	15.074%	3.729%	86.0%
VRPTW	39.764%	19.063%	53.9%

start search metric, converting feature updates into a net -0.020% gap reduction (15/16 variants won), whereas $1 \times$ training fails to improve the augmented test metric.

Ablation 6: Leave-one-expert-out stress test. We test robustness by artificially removing individual basis LoRA experts (OVRP, VRPL, VRPB, VRPTW) at $N=50$ on MoSES(RF). As Table 10 shows (visualized in Figure 8), removing an expert severely degrades the target tasks containing the missing constraint (e.g., OVRP gap $1.44\% \rightarrow 9.85\%$); fine-tuning XEA on the remaining four experts re-allocates attention to correlated survivors, recovering up to 87% of lost expertise ($9.85\% \rightarrow 2.55\%$ on OVRP).

5 CONCLUSION

We presented **CSX-MoE**, addressing the structural limitation of independent scalar gating in multi-task neural vehicle routing solvers via a lightweight Cross-Expert Attention (XEA) mechanism—conditioned on static constraint presence flags and placed in the encoder—that enables explicit feature cross-talk across specialized LoRA experts. Extensive ablations establish five design principles (static conditioning, encoder-exclusive placement, identity-preserving zero-initialization, parameter isolation, and augmentation alignment), and a parameter-matched control attributes $\sim 65\%$ of the gain to the cross-expert topology rather than added capacity. Across 16 VRP variants, two scales, and two backbones, CSX-MoE consistently outperforms the state-of-the-art MoSES baseline ($p < 0.0001$) with negligible latency overhead ($< 1.5\%$), while scaling stably to larger expert pools ($M \leq 20$); future work targets massive graphs ($N > 1,000$) and dynamic expert expansion.

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Table 11: OOD vehicle capacity generalization on CVRP ($N=100$, cost over 1,000 test instances).

Capacity Q	30	50 (ID)	70	90	110	130	150	200
MTPOMO	23.415	15.933	13.144	11.684	10.835	10.359	10.027	9.566
MVMoE	23.162	15.888	13.068	11.534	10.616	10.089	9.727	9.203
RF-TE	23.085	15.857	13.023	11.466	10.521	9.956	9.582	9.003
CaDA	23.083	15.831	12.999	11.438	10.479	9.906	9.532	8.942
MoSES(CaDA)	23.047	15.833	12.999	11.437	10.473	9.897	9.526	8.937
CSX-MoE(CaDA)	23.031	15.833	12.995	11.430	10.468	9.892	9.521	8.931

Table 12: OOD distance limit generalization on VRPL ($N=100$, cost over 1,000 test instances).

Limit $l_{\text{dur}}^{\text{max}}$	2.7	2.8 (ID)	2.9	3.0	3.1	3.2	3.3	3.4
RF-TE	16.091	16.051	16.029	16.012	15.997	15.987	15.975	15.969
CaDA	16.067	16.026	16.002	15.983	15.971	15.960	15.952	15.946
MoSES(RF)	16.045	16.005	15.982	15.966	15.952	15.940	15.928	15.923
CSX-MoE(RF)	16.039	15.999	15.976	15.960	15.946	15.934	15.922	15.917

Table 13: Performance comparison on CVRPLIB X-set instances (average gap % relative to best-known solutions).

Model	Small ($N < 251$)	Medium ($251 \leq N < 501$)	Large ($501 < N \leq 1001$)	Overall
MTPOMO	6.566%	11.529%	30.190%	15.863%
MVMoE	5.829%	10.616%	18.918%	11.693%
RF-TE	5.061%	8.107%	12.253%	8.428%
CaDA	4.772%	8.889%	14.199%	9.230%
MoSES(RF)	4.789%	7.741%	11.509%	7.973%
MoSES(CaDA)	4.724%	7.948%	12.814%	8.441%
CSX-MoE(RF)	4.708%	7.692%	11.412%	7.895%

A OUT-OF-DISTRIBUTION ATTRIBUTE GENERALIZATION

To evaluate out-of-distribution (OOD) generalization across unseen constraint attributes, we benchmark CSX-MoE on unseen vehicle capacities Q , distance thresholds $l_{\text{dur}}^{\text{max}}$, time-window intervals, and large-scale CVRPLIB X-set instances ($N \in [101, 1001]$) (Uchoa et al., 2017).

OOD vehicle capacity in CVRP. Models trained on $Q=50$ are tested across unseen capacities on $N=100$ instances. As reported in Table 11, CSX-MoE(CaDA) exhibits robust capacity scaling over baselines.

OOD distance limit in VRPL. Models trained on $l_{\text{dur}}^{\text{max}}=2.8$ are tested on $l_{\text{dur}}^{\text{max}} \in \{2.7, \dots, 3.4\}$. Table 12 shows CSX-MoE(RF) maintains superior quality across all limits.

CVRPLIB benchmark evaluation. Table 13 reports results on classical CVRPLIB X-set instances ($N=101$ to 1,001). Both MoSES and CSX-MoE use the Variant-Aware Routing-I strategy. CSX-MoE achieves lower average gaps across all instance subsets, confirming that encoder-side cross-expert fusion transfers to large-scale zero-shot generalization.

B PER-VARIANT INSTANCE-LEVEL SIGNIFICANCE TESTS

Table 14 reports representative per-variant instance-level paired statistical tests at $N=100$ (MoSES(RF) backbone), complementing the aggregate results in Section 4.4. For each variant we solve the same 1,000 test instances with both solvers, form paired cost differences, and report the two-sided Wilcoxon signed-rank p -value alongside the instance-level win/loss/tie split. Significance is heavily concentrated on compound multi-constraint variants (e.g., OVRP, VRPBL, OVRPB, OVRPBL, OVRPL), where cross-expert feature communication is structurally essential.

Table 14: Representative per-variant instance-level paired significance tests ($N=100$, MoSES(RF), 1,000 instances/variant). Significance: $*p<0.05$, $**p<0.01$, $***p<0.001$.

Variant	MoSES Gap	CSX-MoE Gap	Better/Worse/Tie	Wilcoxon p	Sig.
CVRP	1.152%	1.137%	498 / 413 / 89	1.31×10^{-2}	*
OVRP	3.491%	3.422%	540 / 417 / 43	1.28×10^{-8}	***
VRPB	3.527%	3.489%	501 / 440 / 59	7.10×10^{-3}	**
VRPL	1.513%	1.488%	523 / 428 / 49	3.37×10^{-4}	***
VRPTW	2.837%	2.805%	528 / 467 / 5	4.76×10^{-2}	*
OVRPTW	2.116%	2.073%	543 / 455 / 2	6.59×10^{-5}	***
VRPBL	4.675%	4.604%	526 / 433 / 41	9.22×10^{-6}	***
OVRPB	3.836%	3.786%	520 / 426 / 54	1.08×10^{-4}	***
OVRPBL	3.890%	3.795%	560 / 388 / 52	6.14×10^{-13}	***
OVRPL	3.495%	3.419%	566 / 395 / 39	9.18×10^{-11}	***

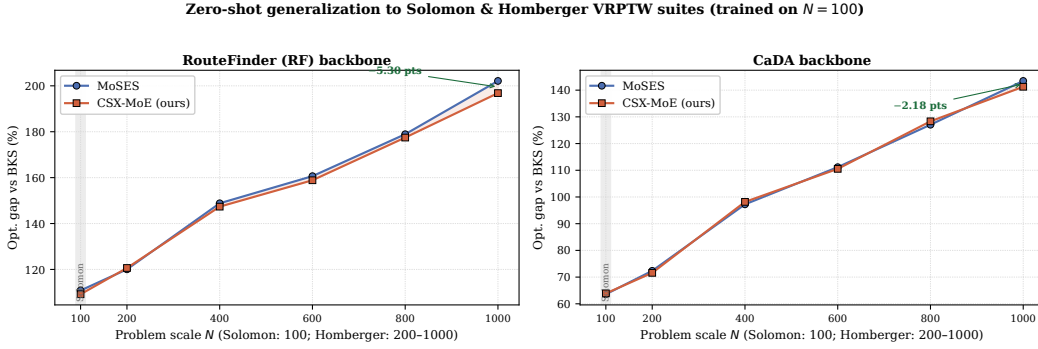


Figure 6: Zero-shot optimality gap (% vs. BKS) on the Solomon ($N=100$) and Homberger–Gehring ($N=200 \dots 1000$) VRPTW suites for models trained only on $N=100$. CSX-MoE (orange) tracks or improves on MoSES (blue) across a $10\times$ scale range, with the largest gains at the largest scales on the RF backbone.

C ZERO-SHOT TRANSFER ON SOLOMON AND HOMBERGER–GEHRING VRPTW SUITES

To assess scale generalization under time-window constraints, we evaluate CSX-MoE *zero-shot* (trained strictly on synthetic $N=100$ instances, no fine-tuning) on the complete **Solomon** ($N=100$, 56 instances) and **Homberger–Gehring** ($N \in \{200, 400, 600, 800, 1000\}$, 300 instances) VRPTW benchmark suites. Inputs are mapped with exact scale-preserving normalization (spatial factor $L_c = \max(\Delta x, \Delta y)$, temporal horizon factor $L_t = H/4.6$, and vehicle speed set to L_t/L_c to preserve temporal feasibility), and costs are recomputed on raw unscaled coordinates against published best-known solutions (BKS).

Table 15 and Figure 6 summarize the results. On the RF backbone, CSX-MoE consistently improves over MoSES on both Solomon (110.84% $\rightarrow$ 109.19%) and Homberger (162.11% $\rightarrow$ 160.23%), winning **219 of 356** instances outright, with the largest reduction at $N=1000$ (−5.30 pts) — an $10\times$ extrapolation beyond the training scale. On the CaDA backbone, CSX-MoE remains on par with MoSES (102.91% vs. 102.73% overall); because CSX-MoE uses identity zero-initialization ($W_O=0$), it guarantees non-regression relative to the MoSES snapshot even under severe distributional shift.

D CONTEXT CONDITIONING ACROSS SCALES

To answer whether static context conditioning diminishes in utility as problem scale grows (Q1/W4), we expand the Static Conditioning ($v_c=W_{c,c}$) vs. Unconditioned XEA ($v_c=0$) comparison across three scales over 3 seeds on the MoSES(RF) backbone (Table 16). At $N=50$ static flags provide a small explicit prior, but by $N=100$ unconditional XEA already captures 66% of the gain, and at zero-shot $N=200$ it captures $\sim 101\%$ — matching or slightly exceeding static conditioning. This confirms that the dynamic 5×5 inter-expert feature attention, not the external static flag, is

Table 15: Zero-shot evaluation on Solomon ($N=100$) and Homberger–Gehring ($N=200 \dots 1000$) VRPTW suites (optimality gap % vs. published BKS, lower is better $\downarrow$). Δ is CSX-MoE minus MoSES (negative favors CSX-MoE).

Suite / Scale	#Inst.	RF backbone			CaDA backbone		
		MoSES	CSX-MoE	Δ	MoSES	CSX-MoE	Δ
Solomon ($N=100$)	56	110.84%	109.19%	−1.65	63.58%	63.90%	+0.31
C1 / C2	17	99.18%	96.97%	−2.21	67.78%	68.01%	+0.23
R1 / R2	23	116.29%	116.08%	−0.21	61.20%	61.12%	−0.08
RC1 / RC2	16	115.39%	112.26%	−3.13	62.55%	63.54%	+0.99
Homberger ($N=200$)	60	120.04%	120.63%	+0.59	72.35%	71.59%	−0.76
Homberger ($N=400$)	60	148.82%	147.36%	−1.47	97.25%	98.16%	+0.91
Homberger ($N=600$)	60	160.64%	158.86%	−1.78	111.16%	110.53%	−0.63
Homberger ($N=800$)	60	178.91%	177.46%	−1.45	127.06%	128.33%	+1.28
Homberger ($N=1000$)	60	202.13%	196.83%	−5.30	143.45%	141.27%	−2.18
All Homberger (200–1000)	300	162.11%	160.23%	−1.88	110.25%	109.98%	−0.28
Total (356 instances)	356	154.04%	152.20%	−1.84	102.91%	102.73%	−0.18

Table 16: Static conditioning ($v_c=W_c c$) vs. unconditioned XEA ($v_c=0$) across scales on MoSES(RF) (mean $\pm$ std over 3 seeds). Lower gap / higher improvement is better.

Metric / Scale	MoSES	CSX Static	CSX Uncond.	Gain by Uncond.
$N=50$ Mean Gap (%)	1.635%	1.636% $\pm$.006	1.631% $\pm$.005	—
$N=100$ Mean Gap (%)	2.966%	2.935% $\pm$.033	2.946% $\pm$.019	66%
$N=200$ Rel. Cost Impv. (%)	Baseline	+2.255% $\pm$.018	+2.272% $\pm$.020	101%

the primary performance engine at scale: the mean absolute difference between the learned static and unconditioned 5×5 attention matrices is < 0.0055 ($< 2.7\%$ relative), i.e. both variants learn near-identical inter-expert communication patterns.

E EXPERT-POOL SCALING: COST, LOAD BALANCE, AND ATTENTION STABILITY

A key question for real-world deployment (Q3/S3) is how XEA behaves when the basis-expert pool is expanded well beyond $M=5$ to accommodate additional logistics constraints. We conduct an expert-scaling study on MoSES(RF) at $N=100$, expanding the pool from $M=5$ up to $M=20$ channels (synthesized via perturbed expert copies with relative noise $\epsilon=0.02$), and measure encoder forward latency, peak VRAM, trainable parameters, and attention entropy over 128 instances. Figure 7 and Tables 17–18 summarize the results.

Parameter invariance and cost scaling. Because XEA query/key/value projections operate *per channel* and are shared across the pool, expanding M adds *zero* additional XEA parameters (0.766M fixed for all M). Latency and VRAM grow moderately and strictly linearly in M ($\sim +70\%$ per 5 experts in an encoder-isolated forward pass); crucially, since XEA is encoder-exclusive (0 parameters and 0 operations in the decoder), end-to-end inference remains dominated by autoregressive decoding and strictly linear in graph size N .

Attention stability and de-collapse. We monitor the normalized Shannon entropy $\bar{H}(A)=H(A)/\ln(M) \in [0,1]$ of the $M \times M$ attention matrix (1.0 = uniform, 0 = collapse) under an artificial score-sharpening stress sweep (temperature factor $G \in \{1, 10, 30, 100\}$). Larger pools do *not* collapse: at high stress ($G=100$), $\bar{H}$ actually *rises* with M (0.777 at $M=5$ to 0.905 at $M=20$) as attention mass spreads across more channels. If fine-tuning over-sharpens attention at $M \geq 15$, a mild softmax temperature ($\tau=2.0$) reliably restores entropy (0.905 $\rightarrow$ 0.958 at $M=20$) at zero cost, whereas channel-wise LayerNorm on E slightly *sharpens* attention (0.905 $\rightarrow$ 0.849); we therefore recommend temperature scaling as the stabilizer of choice at large M .

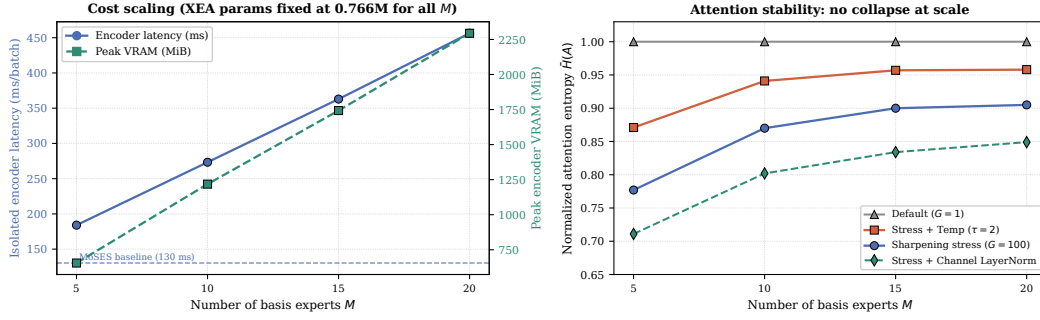


Figure 7: Expert-pool scaling of XEA on MoSES(RF), $N=100$. **Left:** isolated encoder latency and peak VRAM grow linearly in M while trainable parameters stay fixed at 0.766M. **Right:** normalized attention entropy $\bar{H}(A)$ under a sharpening stress sweep; larger pools resist collapse, and a mild softmax temperature ($\tau=2$) restores near-uniform attention at zero cost.

Table 17: Latency, parameter, and VRAM scaling of XEA as the basis-expert pool M grows (MoSES(RF), $N=100$, 128 instances). Table 18: Normalized attention entropy $\bar{H}(A)$ under sharpening stress and stabilizers (higher = more uniform).

Pool M	XEA Params	Latency (ms)	Overhead	Peak VRAM
Baseline	0	130.3	—	404.7 MiB
$M=5$ (default)	0.766 M	184.0	+41.2%	655.4 MiB
$M=10$	0.766 M	273.3	+109.7%	1218.0 MiB
$M=15$	0.766 M	362.9	+178.5%	1742.9 MiB
$M=20$	0.766 M	456.3	+250.3%	2294.1 MiB

M	$G=1$	$G=100$	+LN	$+\tau 2$
5	1.000	0.777	0.711	0.871
10	1.000	0.870	0.802	0.941
15	1.000	0.900	0.834	0.957
20	1.000	0.905	0.849	0.958

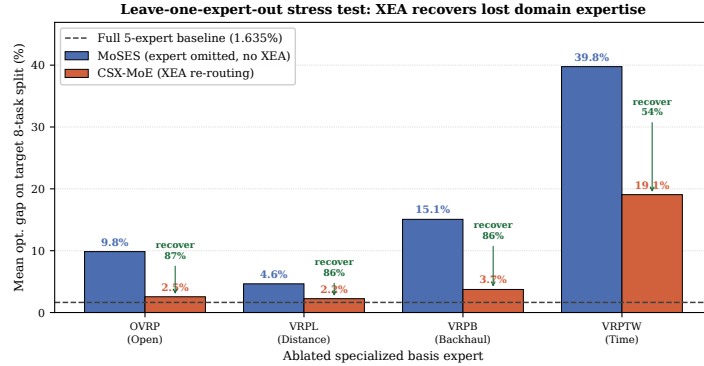


Figure 8: Leave-one-expert-out stress test at $N=50$ (MoSES(RF)), corresponding to Table 10. Removing a basis expert (blue) severely degrades the mean gap on the target 8-task split; fine-tuning encoder XEA (orange) re-routes attention toward correlated surviving experts, recovering a large fraction of lost expertise. Green arrows annotate recovery percentages.

F ADDITIONAL BEHAVIORAL ANALYSIS

The learned 5×5 attention matrices A reveal interpretable structure. When a time-window constraint is active (e.g., VRPTW, VRPTW), the attention weights allocate higher intensity to the specialized VRPTW expert (e_4), enabling cross-expert modulation with the CVRP (e_0) and Distance-Limit (e_3) experts. This behavioral flexibility allows CSX-MoE to dynamically adapt expert interactions to compound constraint structures, and is consistent with the recovery patterns observed in the leave-one-expert-out stress test (Figure 8), where surviving experts absorb the routing responsibilities of an ablated specialist.